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1) Given:

The number of books in a particular section at the start of the academic year 2025 is 2000.

$\therefore$ Initial count is $b(0) = 2000$

It is also given that at the start of any subsequent week, the number of books increase by 1.5% compared to what is exactly one week earlier.

$\therefore$ % increase will be 0.015 at the start of every subsequent week.

a) Find a function $b: [0, \infty)$

Since the number of books increases every week by 1.5% we can assume that the function b increases exponentially at the rate of time in weeks.

$b =$ no. of books

$t =$ time.

$$b(t) = b_0 (1 + 0.015)^t$$

$b_0 = 2000$ since initially the number of books were 2000

$\therefore b(t) = 2000(1.015)^t$	for all $t \in \mathbb{R} \rightarrow (0, \infty)$
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b) Explain why b^{-1} exists.

$$b(t) = 2000(1.015)^t$$

Here we can see that for any week t , the number of books will be unique.

$\therefore$ For all x there is a unique y value

$\therefore$ The function is one to one

Also there are no two ~~the~~ weeks which ~~is~~ will have same no. of books at increasing continuously. no. of books is

$\therefore$ All x_1, x_2 are unique $x_1 \neq x_2$ ✓

$\therefore$ The function is bijective.

2.1

$\therefore b(t)$ has an inverse function.

c) Find b^{-1} for $b(t)$

$$b(t) = 2000(1.015)^t$$

$$b = 2000(1.015)^t$$

Take $\ln$ on both sides.

$$\ln(b) = \ln(2000) + t \ln(1.015)$$

$$t \ln(1.015) = \ln(b) - \ln(2000)$$

2.2

$$t = \frac{\ln(b) - \ln(2000)}{\ln(1.015)}$$
 ✓

d) The function b^{-1} ~~is~~

$$\text{ie } b^{-1} = t = \frac{\ln(b) - \ln(2000)}{\ln(1.015)}$$

Here the function takes the number of books as input and returns the time at which the number of books was present. ✓

e) Find $b'(t)$

$$b(t) = 2000(1.015)^t$$

$$b'(t) = 2000(1.015)^t \times \ln(1.015)$$

$$b'(t) = b(t) \ln(1.015) \quad (\because b(t) = 2000(1.015)^t)$$

$b'(t)$ represents at what rate the books keep increasing at time t . ✓

f) Find the number of books after 52 weeks.

$$\text{wkt } b(t) = 2000(1.015)^t$$

Given $t = 52$ weeks.

$$\therefore b(52) = 2000(1.015)^{52}$$

using calculator

I found $(1.015)^{52} \approx 2.168$

$$\therefore b(52) = 2000 \times 2.168$$

using calculator

$$b(52) \approx 4336$$

$\therefore$ At week 52, the number of books is approximately 4336 books. ✓

2) Given Domain $[0, 4]$

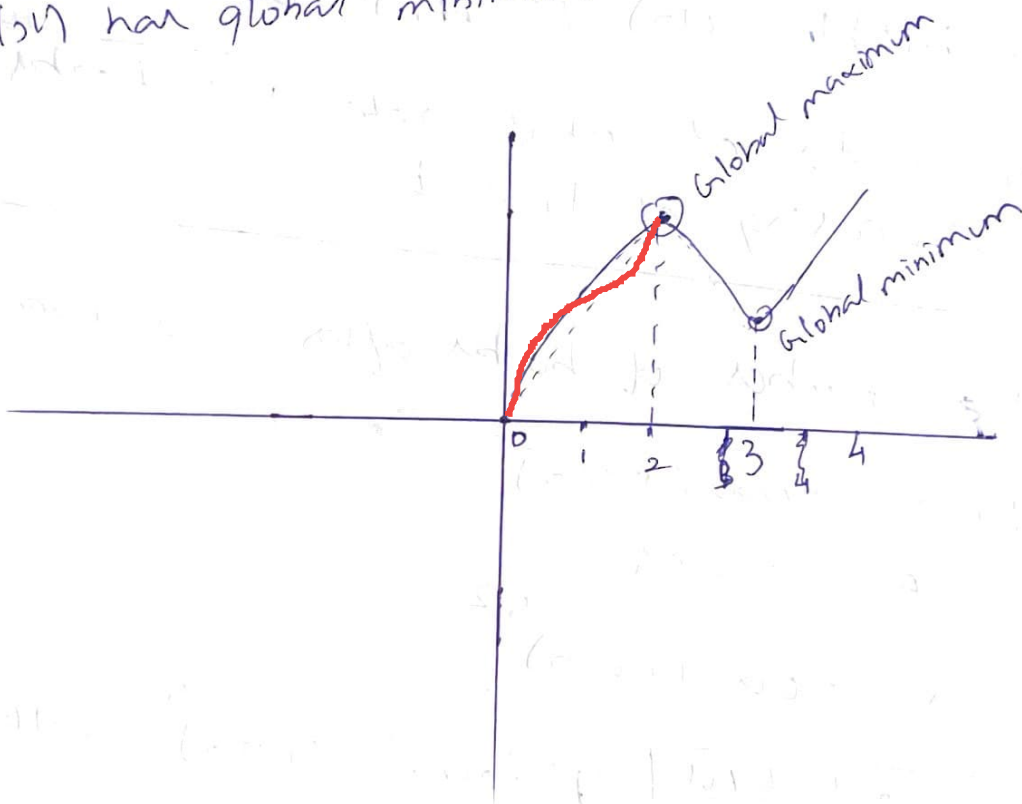
x ranges from 0 till 4.

a)

i) $f'(x) > 0$ on $[0, 2)$ and $(3, 4]$: $f(x)$ is increasing
 $f'(x) < 0$ on $(2, 3)$: $f(x)$ is decreasing on these intervals.

ii) $f''(x) \leq 0$ on $(0, 1)$ $f(x)$ is a concave function.
 $f''(x) \geq 0$ on $(1, 2)$ $f(x)$ is a convex function.

iii) $f(x)$ has global maximum at $x = 2$
 $f(x)$ has global minimum at $x = 3$

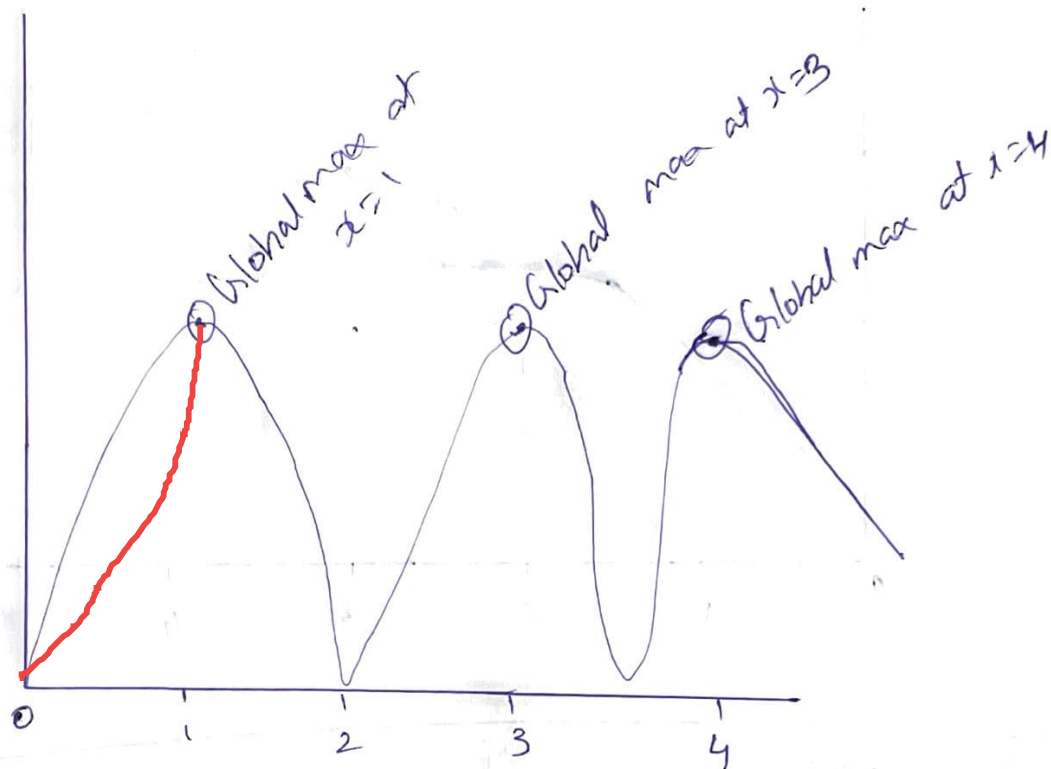


b) $g(x)$ such that

i) $g''(x) \geq 0$ on $[0, 1]$ and $[3, 4]$

is a convex function.

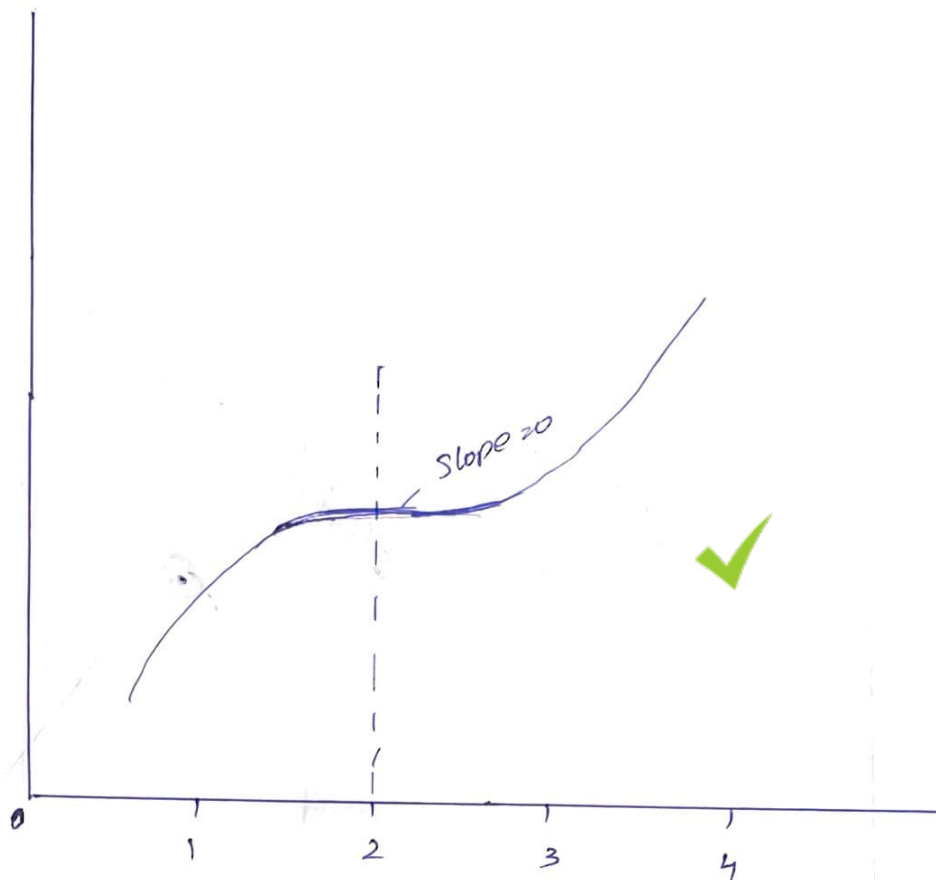
ii) $g(x)$ has global maximum at $x=1$ $x=3$ + $x=4$.



$$c) h'(2) = 0$$

$$h'(x) \geq 0 \text{ for all } x \in [0, 4]$$

$h(x)$ does not have a local maximum + local minimum.



$$3) f(x) = mx + c$$

$$RSS = \sum_{i=1}^n (y_i - f(x_i))^2$$

$$RSS = \sum_{i=1}^n (y_i - (mx_i + c))^2$$

$$= \sum_{i=1}^n (y_i^2 - mx_i^2 + c^2 + 2mx_i c)$$

$$= \sum_{i=1}^n (y_i^2 - mx_i^2 + c^2 + 2mx_i c)$$

$$3) f(x) = mx + c$$

$$RSS = \sum_{i=1}^n (y_i - b(x_i))^2$$

a) General formula for m:

$$RSS = \sum_{i=1}^n (y_i - (mx_i + c))^2$$

$$= \sum_{i=1}^n (y_i - mx_i - c)^2$$

Differentiating both sides:

$$\sum_{i=1}^n 2(y_i - mx_i - c)(-x_i) = 0$$

$$-2 \sum_{i=1}^n (x_i (y_i - mx_i - c)) = 0$$

$$\sum x_i y_i - m \sum x_i^2 - c \sum x_i = 0$$

$$-m \sum x_i^2 = c \sum x_i - \sum x_i y_i$$

$$m \sum x_i^2 = \sum x_i y_i - c \sum x_i$$

$$m = \frac{\sum_{i=1}^n x_i y_i - c \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2}$$



$$b) (x_i, y_i) = (1, 2) (3, 3) (4, 5)$$

$$\boxed{b(x) = mx + 1}$$

from (a), we can get

$$m = \frac{\sum_{i=1}^n x_i y_i - c \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2}$$

Now we find each term to substitute in m.

$$\begin{aligned} \sum x_i y_i &= (1 \times 2) + (3 \times 3) + (4 \times 5) \\ &= 2 + 9 + 20 \\ &= 31 \end{aligned}$$

Here, constant term $c = 1$. $\left(\because b(x) = mx + 1 \right)$
 $c = 1$

$$\sum x_i = 1 + 3 + 4 = 8$$

$$\begin{aligned} \sum x_i^2 &= 1^2 + 3^2 + 4^2 \\ &= 1 + 9 + 16 \\ &= 26. \end{aligned}$$

Find m:

$$m = \frac{31 - 1(8)}{26} = \frac{31 - 8}{26} = \frac{23}{26}$$

using calculator

$$\boxed{m = 0.8846}$$



Substitute m value in $f(x)$

$$m = 0.8846 \quad f(x) = mx + 1$$

$$\therefore f(x) = 0.8846x + 1$$

To find y values, we substitute all x values in $f(x)$ because $f(x) = y$.

$$f(x_1) = f(1) = 0.8846(1) + 1 = 1.8846$$

$$f(x_2) = f(3) = 0.8846(3) + 1 = 2.6538 + 1 = 3.6538$$

(using calculator)

$$f(x_3) = f(4) = 0.8846(4) + 1 = 3.5384 + 1 = 4.5384$$

(using calculator)

To find RSS error,

$$RSS = (y_1 - f(x_1))^2 + (y_2 - f(x_2))^2 + (y_3 - f(x_3))^2$$

$$\Rightarrow (2 - 1.8846)^2 + (3 - 3.6538)^2 + (5 - 4.5384)^2$$

$$= (0.1154)^2 + (0.6538)^2 + (0.4616)^2$$

$$= \text{Using calculator}$$

$$= 0.0133 + 0.4274 + 0.2130$$

$$= 0.6537 \quad (\text{using calculator})$$

$$\boxed{RSS \text{ error} = 0.6537}$$



$$c) \log_e y = \ln(y).$$

$$\ln(y) = mx + 0.4.$$

$$\text{Given } (x, y) = (1, 2) (3, 3) (4, 5).$$

To find $\ln(y)$, use natural log calculator

$$\ln(2) \approx 0.6931$$

$$\ln(3) = 1.0986$$

$$\ln(5) = 1.6094$$

$$\text{Transformed set } (x_i, \ln(y_i)) = (1, 0.6931) (3, 1.0986) (4, 1.6094)$$

$$\text{from (a), w.k.T } m = \frac{\sum_{i=1}^n x_i y_i - c \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2}$$

$$\text{Sub } y_i = \ln(y_i)$$

$$\Rightarrow m = \frac{\sum x_i \ln(y_i) - c \sum x_i}{\sum x_i^2}$$

$$= \frac{\sum x_i (\ln(y_i) - c)}{\sum x_i^2}$$

$$m = \frac{\sum x_i (\ln(y_i) - c)}{\sum x_i^2}$$

To find m :

$$m = \frac{\sum x_i \ln y_i - (\sum x_i)(\sum \ln y_i)}{\sum x_i^2}$$

$$\sum x_i = 1 + 3 + 4 = 8$$

$$\sum x_i^2 = 1^2 + 3^2 + 4^2 = 1 + 9 + 16 = 26$$

$$\begin{aligned}\sum x_i \ln y_i &= (1 \times 0.6931) + (3 \times 1.0986) + (4 \times 1.6094) \\ &= 0.6931 + 3.2958 + 6.4376 = 10.4265\end{aligned}$$

$$\begin{aligned}\sum x_i c &= 8(0.4) \quad (\because c = 0.4) \\ &= 3.2\end{aligned}$$

$$m = \frac{\sum x_i \ln y_i - \sum x_i c}{\sum x_i^2} = \frac{10.4265 - 3.2}{26}$$

$$= \frac{7.2265}{26} = 0.2779$$

$$= 0.2779$$

$$m = 0.2779$$

$$\ln(y) = mx + 0.4$$

To find y , we need to raise both sides to exponentiation.

$$e^{\ln(y)} = e^{mx + 0.4}$$

$$y = e^{0.2779x} + e^{0.4}$$



~~4-14~~

Using calculator,

$$e^{0.4} = 1.4918$$

$$\therefore y = 1.4918 e^{0.2779x}$$

$$q(x) = 1.4918 e^{0.2779x}$$

To compute RSS

$$\begin{aligned} q(1) &= 1.4918 e^{0.2779(1)} \\ &= 1.4918 \times e^{0.2779} \end{aligned}$$

using calculator,

$$e^{0.2779} = 1.3203$$

$$\therefore q(1) = 1.4918 \times 1.3203$$

$$q(1) = 1.9696$$

$$\begin{aligned} q(3) &= 1.4918 e^{0.2779(3)} \\ &= 1.4918 \times e^{0.8337} \end{aligned}$$

using calculator,

$$e^{0.8337} = 2.3018$$

$$\therefore q(3) = 1.4918 \times 2.3018$$

$$q(3) = 3.4338$$

$$\begin{aligned} q(4) &= 1.4918 \times e^{0.2779(4)} \\ &= 1.4918 \times e^{1.1116} \end{aligned}$$

$$= 1.4918 \times 3.0392$$

$$q(4) = 4.5338$$

$$RSS = (y_1 - q(1))^2 + (y_2 - q(3))^2 + (y_3 - q(4))^2$$

$$= (2 - 1.9696)^2 + (3 - 3.4338)^2 + (5 - 4.5338)^2$$

using calculator

$$RSS = 0.4064$$



d) $y = Ax^b$

Take natural log on both sides.

$$\ln y = \ln A + b \ln x$$

$$\ln y = b \ln x + \ln A$$

wkt $y = mx + c$

$$y = \ln(y)$$

$$m = b$$

$$x = \ln x$$

$$c = \ln(A)$$

From Qn, wkt intercept is 0.6

$$c = \ln A = 0.6$$

$$\ln y = b \ln x + 0.6$$



$$\text{WKT } m = \frac{\sum x_i y_i - c \sum x_i}{\sum x_i^2} = b.$$

Sub
 $x = \ln x_i$ $y = \ln y_i$

$$\Rightarrow m = b = \frac{\sum (\ln x_i \ln y_i) - c \sum \ln x_i}{\sum (\ln x_i)^2}$$

$$\underline{\ln(x_i)}$$

using calculator

$$\ln(1) = 0$$

$$\ln(3) = 1.0986$$

$$\ln(4) = 1.3862$$

$$\underline{\ln(y_i)}$$

$$\ln(2) = 0.6931$$

$$\ln(3) = 1.0986$$

$$\ln(5) = 1.6094$$

$$\therefore \sum \ln x_i = 0 + 1.0986 + 1.3862$$

$$\boxed{\sum \ln x_i = 2.4848}$$

$$\sum \ln y_i = 0.6931 + 1.0986 + 1.6094$$

$$\boxed{\sum \ln y_i = 3.4011}$$

$$\sum x_i^2 = 0^2 + (1.0986)^2 + (1.3862)^2$$

$$\boxed{\sum \ln x_i^2 = 3.1284}$$

$$\sum \ln x \ln y = (0 \times 0.6931) + (1 - 0.986 \times 1.0986)$$

$$+ (1.3862 \times 1.6094)$$

$$\sum \ln x \ln y = 3.4378$$

$$\therefore m = b = 3.4378 - 0.6 (2.4848)$$

$$3.1284$$

$$m = b = 0.6223$$

$$\text{w/ct } \ln A = 0.6$$

raise to the power of e on both sides
to find A .

$$A e^{\ln A} = e^{0.6}$$

$$A = e^{0.6}$$

$$A = 1.8221$$

$$\therefore y = 1.8221 x^{0.6223}$$

$$y(x) = 1.8221 x^{0.6223}$$



To compute RSS, $RSS = \sum_{i=1}^n y_i - b(x_i)]^2$

~~y(1)~~ = $1.8221(1)^{0.6223} = 1.8221$

$y(3) = 1.8221(3)^{0.6223} = 3.6098$

$y(4) = 1.8221(4)^{0.6223} = 4.3175$

$\therefore RSS = (2 - 1.8221)^2 + (3 - 3.6098)^2 + (5 - 4.3175)^2$

$RSS = 0.8693$

using calculator



e) Based on the previous parts,

linear model $RSS = 0.6537$

exponential model $RSS = 0.4064$

power law model $RSS = 0.8693$

from the above 3 RSS values, we can notice that exponential model has the least RSS error of 0.4064. Therefore exponential model is the best fit for the given data.



Index of comments

- 2.1 this implies that the function is injective, but not necessarily surjective
- 2.2 domain not specified
- 7.1 we also need to check $RSS'' > 0$